ALGORITHM

Kth Largest Element in a Binary Search Tree (BST) — Modified Version

1. Start the program.
2. Read the number of elements n. If n is less than or equal to zero, display an error message and stop.
3. Read n distinct integer values and insert each value into the Binary Search Tree.
4. Ask the user how many K values they want to test.
5. For each query, read the value of K.
6. If K is less than 1 or greater than n, display “Invalid value of K” and continue to the next query.
7. For a valid K, initialise the visited-node counter to zero and begin reverse inorder traversal.
8. Visit the right subtree first, then the current node, and finally the left subtree (Right → Root → Left).
9. Display each visited node and increase the counter by one.
10. When the counter reaches K, store the current node as the Kth largest element and stop that query’s traversal.
11. Display the Kth largest element, the nodes visited, and the total number of nodes visited.
12. After all queries, compare the visited-node counts for the valid K values.
13. Stop the program.